

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Sixth Semester of B. Tech IT Examination****May 2022****IT355 Software Engineering****Date: 10.05.2022, Tuesday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Do as directed.****[07]**

(i) Match the following:

[01]

P. Waterfall Model

1. Specification can be developed incrementally

Q. Evolutionary Model

2. Requirements Compromises are inevitable

R. Component Based Software Engg.

3. Explicit Recognition of risk

S. Spiral Development

4. Inflexible Partition of Project in to stages

(A) P-1, Q-2, R-3, S-4

(C) P-4, Q-2, R-1, S-3

(B) P-4, Q-1, R-2, S-3

(D) P-3, Q-1, R-2, S-4

(ii) Arrange the following activities for making a software product by ascending order.

[01]

I. Design strategy

II. Transformation into product

III. Implementation

IV. Requirement gathering

(A) 4,1,3,2

(C) 1,2,3,4

(B) 4,3,2,1

(D) 1,4,2,3

(iii) _____ is a software developing team has a defined leader who coordinates specific tasks and secondary leaders that have responsibility for sub tasks.

[01]

(A) Democratic Decentralized(DD)

(B) Controlled Decentralized(CD)

(C) Controlled Centralized(CC)

(D) None of the mentioned

(iv) Which of the following term is best defined by the statement: "There will be a change of organizational management with different priorities."?

[01]

(A) Staff Turnover

(B) Technology Change

(C) Management Change

(D) Product Competition

(v) Explain which process model is most suitable for the following definition and justify it. [03]

1. A compiler for a new language.
2. Event Management System.
3. Chess.

Q-2.a Compute the function points for the following data set: Inputs =8, Outputs= 12, [04]
Inquiries = 4, Logical files = 41, Interfaces =1 and $\sum F_i = 41$. It is given that the complexity weighting factors for I, O, E, F and N are 4, 5, 4, 10 and 7, respectively. Suppose that program need 70 LOC per FP. Find out size of complete project.

OR

Q-2.a Consider a software project using embedded mode (with Type basic COCOMO model) [04]
with 40,000 line of code (coefficient value of $a=3.6$, $b=1.20$, $c=2.5$ and $d=0.3$).
1) Find effort estimation
2) Find Duration estimation
3) Find Persons estimation

Q-2.b Answer any TWO questions. [10]

- (i) Explain risk management process with suitable diagram in brief. List types of risk with example.
- (ii) What is management spectrum? Describe the four P's briefly.
- (iii) Define the roles in Scrum? Explain Agile software development process in detail.

Q-3.a List various requirement gathering techniques and explain any two in detail. [04]

OR

Q-3.a Illustrate Boehm's W5HH Principle based on online food delivery app like Zomato. [04]

Q-3.b Answer any TWO questions. [10]

- (i) Draw neat sketch of spiral model and justify why it is considered as meta model.
- (ii) Discuss umbrella activities and its role in software development life cycle (SDLC).
- (iii) What are the characteristics of software? Explain hardware and software failure curve in detail.

SECTION - II

Q - 4 Do as directed. [07]

- (i) Which of the following is not a maturity level in CMM? [01]
(A) Managed (B) Repeatable
(C) Design (D) Optimized

- (ii) A program validates numeric fields as follows: values less than 10 are rejected, values [01]
between 10 and 21 are accepted, values greater than or equal to 22 are rejected. Which
of the following input values cover all of the equivalence partitions?
(A) 10,11,21 (B) 03,10,22
(C) 03,20,21 (D) 10,21,22

- (iii) _____ specification is also known as SRS document. [01]
 (A) Black-box Testing (B) Grey-box Testing
 (C) White-box Testing (D) None of the mentioned
- (iv) What is verification and Validation? How they are different from each other? [02]
 (v) Distinguish between alpha and beta testing. [02]
- Q-5.a Draw UML diagrams (Use Case, Activity and Sequence) for the online ticket booking system for IPL2022 Twenty 20 Tournament. [07]
- Q-5.b i. Define software maintenance and discuss its types in detail. [04]
 ii. Differentiate throw-away v/s evolutionary model. [03]
- OR**
- Q-5.b i. Consider an application that requires two integer inputs x and y. Each of these inputs is expected to lie in the following ranges: $8 \leq x \leq 13$ and $10 \leq y \leq 19$. Mention different test cases using boundary value analysis. [04]
 ii. Design test case for login functionality (Invalid login, Forget Password and Lock account after 3 attempts) [03]
- Q-6.a Draw context and data flow diagram (DFD) up to level 1 for Online Job Portal. [04]
- Q-6.b Answer any TWO questions. [10]
 (i) Draw neat sketch of CMM levels and mention different process areas of CMM.
 (ii) List seven software testing principles and explain any three principles in detail.
 (iii) Blackbox testing is also called functional testing. Justify and also explain equivalence class partitioning approach of black box testing with example.
